

E15 — Scalability: Muon vs SGD at Larger Dimensions
Experiment Overview This experiment addresses a critical question for practical deployment: **Does Muon's $O(d^3)$ SVD overhead cancel its convergence advantage as problem dimension increases?** As dimension d grows:- **SGD per-step cost:** $O(md^2)$ — scales quadratically- **Muon per-step cost:** $O(md^2 + d^3)$ — the d^3 SVD term dominates at large d - If Muon's iteration advantage is constant but its per-step cost grows faster, the total FLOPs advantage may shrink or reverse. **Tested Dimensions:** d in $\{100, 200\}$ ($d=500$ skipped per engineering compromises)- At $d=100$: SVD cost = $12 * 100^3 = 1.2e7$ FLOPs- At $d=200$: SVD cost = $12 * 200^3 = 9.6e7$ FLOPs (8x more) **Configuration:** $r=5$, 1000 iterations, 3 seeds (reduced for runtime)

Scientific Question **Hypothesis:** As d increases, Muon's SVD overhead grows as $O(d^3)$ while SGD's cost grows as $O(d^2)$. Therefore, the total FLOPs efficiency ratio $\rho_F = F_{\text{SGD}} / F_{\text{Muon}}$ will decrease with d , potentially crossing below 1. **Key Metrics:-** K_{epsilon} : Iterations to convergence- time_s : Wall-clock time- F_{eps} : Total FLOPs- $\rho_F = F_{\text{SGD}} / F_{\text{Muon}}$: Efficiency ratio (>1 means Muon is more FLOP-efficient)- I_{conv} : Convergence flag **Analysis Goals:** 1. Track K_{epsilon} vs d for both algorithms 2. Track time_s

vs d3. Track F_{eps} vs d4. Compute and plot ρ_F vs d

Experimental Design | Parameter | Value | |-----
 ---|-----| | Problem | Matrix Sensing (MS) | |
 Algorithms | Muon-Exact, SGD | | Dimensions d |
 100, 200 | | Target rank r | 5 | | Measurements m |
 | $2dr$ | | Learning rate | 0.01 | | Max iterations |
 1000 | | Random seeds | 3 (0-2) | Reduced seed
 count (3 vs 10) due to substantially longer
 runtime at d=200.

Data Loading & Inspection

```
In [ ]: import pandas as pdimport numpy as npimport matplotlib.pyplot as pltimport s
```

Exploratory Data Analysis### Summary by Algorithm and Dimension

```
In [ ]: summary = df.groupby(['algo', 'd']).agg(n=('seed', 'count'), K_eps_me
```

Critical Observation: Convergence Failure**Muon-Exact fails to converge within 1000 iterations at both d=100 and d=200** ($I_{\text{conv}}=0$ for all seeds, K_{epsilon} capped at 1001). Meanwhile, **SGD converges successfully** at both dimensions ($K_{\text{epsilon}} \sim 49$ at d=100, ~ 53 at d=200). This is a striking result: despite Muon's advantage at d=50, the reduced iteration budget (1000 vs 2000 in E11-E14) and larger dimension cause Muon to fail entirely.

Comparative Analysis### Compute Efficiency Ratio ρ_F

```
In [ ]: # Compute rho_F = F_SGD / F_Muon for each dimensionprint('='*70)print('EFFIC
```

Visualizations

Plot 1: K_{epsilon} vs Dimension d

```
In [ ]: fig, ax = plt.subplots(figsize=(10, 7)) dims = sorted(df.d.unique()) muon_k =
```

Plot 2: Wall-clock Time vs Dimension

```
In [ ]: fig, ax = plt.subplots(figsize=(10, 7)) muon_t = [df[(df['algo'] == 'Muon-Exa
```

Plot 3: FLOPs vs Dimension

```
In [ ]: fig, ax = plt.subplots(figsize=(10, 7)) muon_f = [df[(df['algo'] == 'Muon-Exa
```

Plot 4: Efficiency Ratio rho_F vs Dimension

```
In [ ]: fig, ax = plt.subplots(figsize=(10, 7)) rho_values = [] for d in dims: muor
```

Statistical Tests### Comparing Muon vs SGD at each dimension

```
In [ ]: from scipy.stats import ttest_ind, mannwhitneyu print('='*80) print('STATISTIC
```

Conclusions & Interpretation### Key Findings

1. Muon fails to converge at larger d with reduced budget:

At d=100 and d=200 with 1000 iterations, Muon-Exact does not reach epsilon=0.01 within the budget (K_epsilon = 1001 for all seeds). In contrast, SGD converges consistently (K_epsilon ~49-53).

2. SGD's convergence is robust to dimension scaling:

SGD's K_epsilon grows only modestly with d (~49 at d=100, ~53 at d=200), demonstrating robust scaling behavior.

3. Wall-clock time divergence:

At d=200, Muon's wall-clock time (~1475s) approaches SGD's time (~1490s), but Muon does not converge while SGD does. The SVD overhead is consuming time without delivering convergence.

4. FLOP efficiency ratio rho_F < 1:

At both d=100 and d=200, rho_F = F_SGD / F_Muon < 1, meaning SGD uses fewer total FLOPs to reach convergence. At d=100, rho_F ~0.029; at d=200, rho_F ~0.032.

5. The O(d^3) overhead is the critical bottleneck:

While Muon's spectral

normalization helps at small d with generous iteration budgets, the SVD cost becomes prohibitive as d increases, especially when convergence requires many iterations.###

Implications- **Critical scalability concern:** Muon's SVD overhead poses a serious scalability challenge. At $d=200$, it fails to converge in 1000 iterations where SGD succeeds easily.-

Approximate SVD is essential: The results strongly motivate using approximate SVD methods (RandSVD, truncated SVD) to reduce the $O(d^3)$ bottleneck.- **Iteration budget matters:**

Muon may need significantly more iterations at larger d . With only 1000 iterations, it plateaus.

With 2000 iterations (as in E11-E14), it converges at $d=50$.- **Practical recommendation:** For $d \geq 100$, SGD is the more reliable choice unless Muon

is paired with efficient approximate SVD and a sufficiently large iteration budget.- **Future work:**

Testing Muon-RandSVD at larger d would be valuable to see if the approximation restores scalability.